

AGENEWTECH

H1503RQ

Smart Module Specification_V1.3



Our aim is to provide customers with timely and comprehensive service. For any assistance, please contact our company headquarters:

Wuxi Sunning Technology Co., Ltd. (Abbreviation: Agenew)

No.52, Zone C, Huigu Pioneer Park, Huishan Economic Development Zone, Wuxi

Postal Code: 214174

PhoneNum: +86 0510 8235 0372 Mailbox: Asupport@agenewtech.com

Introduction

AGENEW OFFERS THE DOCUMENT AS A SERVICE TO ITS CUSTOMERS. CUSTOMERS MUST DESIGN THEIR PRODUCTS ACCORDING TO THE SPECIFICATIONS AND PARAMETERS PROVIDED IN THE DOCUMENTS. AGENEW SHALL NOT BE LIABLE FOR ANY DAMAGE CAUSED BY FAILURE TO COMPLY WITH RELEVANT OPERATION OR DESIGN SPECIFICATIONS. AGENEW HAS THE RIGHT TO UPDATE THE DOCUMENT BEFORE IT IS ANNOUNCED.

Disclaimers

AGENEW MAKES EFFORT TO ENSURE THE INTEGRITY, ACCURACY, TIMELINESS OR UTILITY OF THE FUNCTIONS UNDER DEVELOPMENT, BUT DOES NOT EXCLUDE THE POSSIBILITY OF ERRORS OR OMISSIONS OF THE ABOVE FUNCTIONS. UNLESS OTHERWISE SPECIFIED VALID AGREEMENTS, AGENEW MAKES NO IMPLIED OR EXPRESS GUARANTEE FOR THE USE OF FUNCTIONS UNDER DEVELOPMENT. TO THE MAXIMUM EXTENT PERMITTED BY APPLICABLE LAWS, AGENEW SHALL NOT BE LIABLE FOR ANY LOSS OR DAMAGE SUFFERED DUE TO THE USE OF THE FUNCTIONS UNDER DEVELOPMENT, WHETHER SUCH LOSS OR DAMAGE IS FORESEEABLE OR NOT.

Copyright

THE INFORMATION CONTAINED HERE IS PROPRIETARY TECHNICAL INFORMATION OF WUXI SUNNING TECHNOLOGY CO.,LTD. ANY COPIES AND REPRINTS THIS DOCUMENT WITHOUT THE PERMISSION OF AGENEW WILL BEAR LEGAL RESPONSIBILITY.

Copyright © Wuxi Sunning Technology Co., Ltd. 2026. All rights reserved.

About the Document

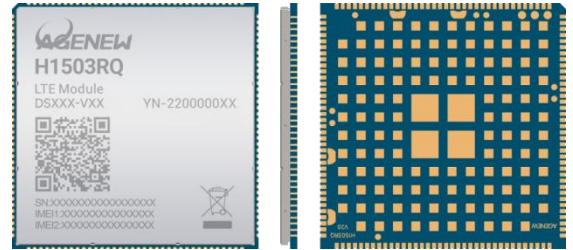
History

Version	Data	Author	Description
-	2025-09-08	Li Ye	Initial
1.0	2025-09-08	Li Ye	Controlled Copy
1.1	2025-09-27	Wu Danyi	Modify frequency band information
1.2	2025-12-15	Wu Danyi	Update display screen resolution information
1.3	2026-03-23	Chen Yun	Change company name

H1503RQ

Multi-network Smart LTE module

Supports Wi-Fi & BT



H1503RQ is a multi-network Smart LTE Cat 4 module based on MediaTek's Octa-core (ARM 2×Cortex-A75 & ARM 6×Cortex-A55) processor launched by Agnew, equipped with Android 14 operating system. Its powerful performance, neon engine processing, 2D/3D graphics acceleration processing capabilities, fully meet the needs of customers for high-speed and multimedia functions in industrial and consumer applications.

H1503RQ supports full coverage of all network systems, WI-FI 802.11a/b/g/n/ac, and BT5.0 near-field wireless communication. In addition, the H1503RQ also has audio and video input and output interfaces and rich GPIO interfaces. It also supports a variety of audio and video codecs. The home screen supports FHD+ 2520×1080 at most. At the same time, H1503RQ supports GPS+GLONASS+BDS+Galileo positioning systems, making its positioning function fast and accurate.

H1503RQ is a patch module with a total of 275 pins, including 146 LCC pins and 129 LGA pins. The size is only 40.5mm × 40.5mm × 2.8mm, which can be embedded in various M2M product applications through pads. H1503RQ integrates rich interfaces (such as LCM, camera, touch screen, microphone, speaker, UART, USB, I2C, SPI, SDIO, and ADC interfaces), greatly expanding its application in the M2M field, making it widely used in smart cash registers, smart POS, tax control machines, security monitoring, vehicle equipment, and high-end information Collection equipment, smart robots, smart homes, smart hardware, industrial smart handheld devices, drones, high-end police or law enforcement equipment, smart intercom devices, smart wearables, vending machines, logistics cabinets and other equipment are used in different industries.



Main Advantages

- ✓ Comprehensive coverage of multi-network standard
- ✓ Support Wi-Fi 802.11a/b/g/n/ac and BT5.0 short-range wireless communication
- ✓ Support GPS/GLONASS/BDS/Galileo
- ✓ Support multiple audio and video codecs
- ✓ The main screen supports up to FHD+ 2520×1080@60fps
- ✓ Support 2K@30fps 1080P@60fps video recording and playback
- ✓ Possess audio and video input and output interfaces and rich GPIO interfaces



Octa-core ARM 2×Cortex-A75
& ARM 6×Cortex-A55



IEEE 802.11
a/b/g/n/ac



Support Android 14



GPS+GLONASS+BDS+Galileo



Powerful multimedia



BT5.0

LTE Cat4		H1503RQ	
Region/Operator	Eurasia/America		
Operating System	Android 14		
Memory	4GB+64GB/6GB+64GB/8GB+64GB		
Module size	40.5 × 40.5 × 2.8		
Packaging	LCC + LGA		
Temperature Range			
Operation Temperature	Normal operating temperature: -20 °C ~ + 80°C		
	Storage temperature range: -40 °C ~ + 90 °C		
Frequency Band Information	H1503RQ-E1	H1503RQ-A1	
LTE-FDD	B1/B2/B3/B4/B5/B7/B8/B20/B28	B2/B4/B5/B7/B12/B13/B17/B25/B26/B66/B71	
LTE-TDD	B38/B40/B41(120M)	B38/B41	
WCDMA	B1/B2/B4/B5/B8	B2/B4/B5	
GSM/EDGE	B2/B3/B5/B8	B2/B5	
WLAN	2.4/5GHz, 802.11 a/b/g/n/ac		
Bluetooth	BT5.0		
GNSS	GPS+GLONASS+BDS+Galileo		
Authentication			
Operator Certification	Undetermined		
Compulsory Certification	Undetermined		
Other Certifications	Undetermined		
Data Rate			
LTE-FDD (Mbps)	150Mbps (DL)/50Mbps (UL)		
LTE-TDD (Mbps)	150Mbps (DL)/50Mbps (UL)		
DC-HSPA+ (Mbps)	42Mbps (DL)/11.2Mbps (UL)		
WCDMA (Kbps)	384Kbps (DL)/384Kbps (UL)		
EDGE (Kbps)	296Kbps (DL)/236.8Kbps (UL)		
GPRS (Kbps)	107Kbps (DL)/85.6Kbps (UL)		
Outstanding characteristics			
Five-Mode	LTE: Release 12 UMTS: Release 9		
Bandwidth（MHz）	1.4/3/5/10/15/20		
Rx-Diversity	●		
DL supports MIMO 2 × 2	●		
SIM Card Detection	●		
Dual SIM Dual Standby	●		
USB Firmware Upgrade	●		
OTA Firmware Upgrade	●		

Remarks: 1. *: In progress

2. ●: Supports

3. Extended operating temperature: If the module works in this temperature range, its RF performance might be worse but it can still work properly.

LTE Cat4	H1503RQ
Electrical specification	
Operating Voltage	3.45~4.4 V, typical 3.8 V
Consumption (typical)	Undetermined
Interface	
LCD	4-lane MIPI_DSI, FHD+ 2520×1080@60fps
Camera	×3(4-lane+2-lane+2-lane) or ×2(4-lane+4-lane), up to 25MP
TP	Capacitive touch screen
Audio	×3 Analog inputs: MIC1, MIC2, HEADMIC (×1 noise cancelling microphone) ×3 Analog outputs: differential linear (requires external audio amplifier), differential receiver and stereo earphone
USB	USB 2.0, Supports OTG, charge
(U)SIM	×2, Supports 1.8/3V (U) SIM card, supports (U) SIM card hot plug detection, dual card dual standby
UART	×2
SD Card	×1
ADC	×2, 1 set general ADC interface, 1 set for battery ID
I2C	×6
SPI	×5
PWRKEY	Internal pull up to VBAT
GPIO	27 GPIO reserved
ANT	Main antenna, Diversity antenna, GNSS antenna, WIFI/BT antenna